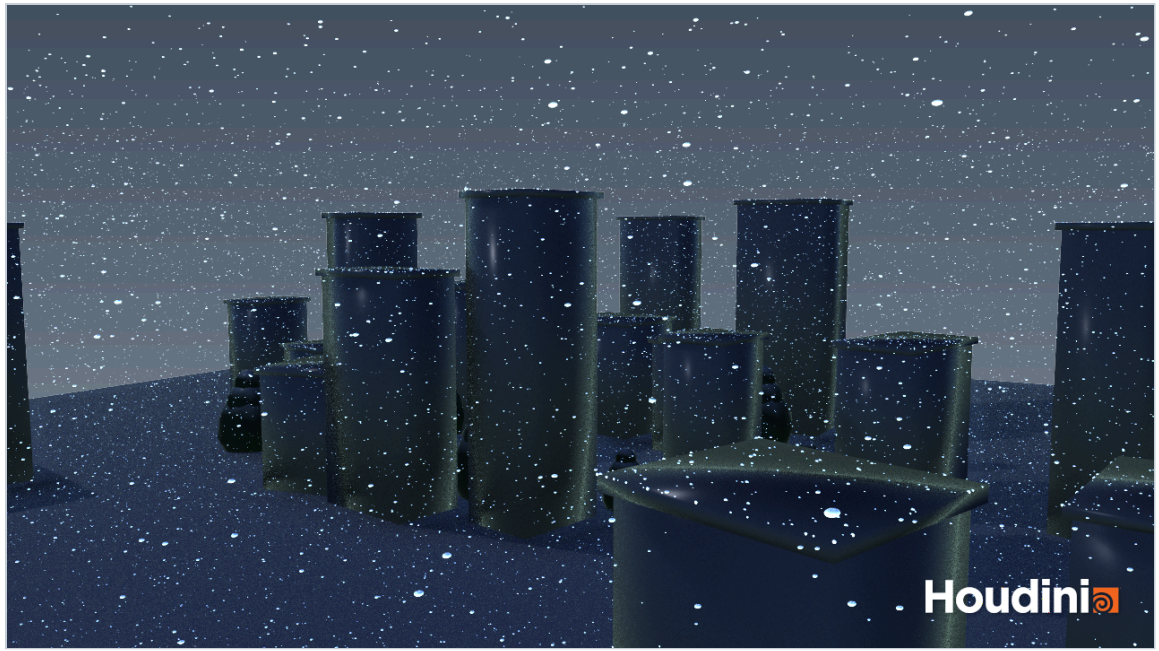
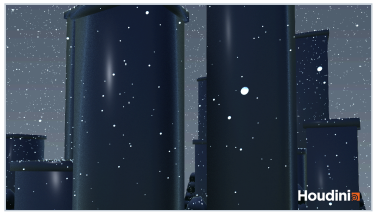
Snowfall over an overcast town / Houdini POP Solver

DURATION	FRAMES	RATE	IMAGE
6.0 s	144	24 fps	1280 × 720









Top: whole shot, frame 480. **Below:** the camera move — wide at frame 337, mid at 400, push-in at 480. Flakes converge on the lens as it travels.

27,850	7,379	2.2 m/s	14 s
FLAKES IN AIR	SETTLED ON SURFACES	TERMINAL FALL	SIM PRE-ROLL

THE SHOT

A slow push-in through a snowstorm over a stylised town. Snow falls at a natural 2.2 m/s terminal velocity, drifts on a light wind and settles on roofs, facades, trees and the ground — about 27,800 flakes in the air at any moment, with roughly 7,400 already stuck to surfaces by the last frame.

UNDER THE HOOD

- **POP Solver** drives the sim; POP Source emits from a 76 × 76 m plane 22 m up.
- **POP Force** gravity −9.81 m/s². **POP Wind** adds a 1.15 m/s breeze plus gusts.
- **POP Drag** is quadratic, tuned to a real snowflake's fall speed.
- **Collision detect + behaviour** against the whole town: flakes stop and stay.
- **POP Kill** retires flakes after 14 s or past a 72 m radius, so the density holds.
- Simulated 14 s of pre-roll, then cached as **144 frames of bgeo.sc**.

IN THIS PACK

Preview loop	MP4 · 2.3 MB
Contact sheet	PNG · every 12th frame
Stills	PNG · 4 frames
One-page brief	this PDF

ON APPROVAL

- **Houdini project** — the full procedural scene, rebuildable from scratch.
- **Geometry cache** — 144 frames of bgeo.sc, compressed, 192 MB.
- **PNG sequence + MP4** — watermark-free 1920 × 1080 flipbook.
- **Source list** — Harvard author–date, verified links.

Licence. This preview is rendered in Houdini Apprentice, which is licensed for non-commercial work and watermarks every frame. The watermark is part of the licence, not the render, and is not removed from previews. On approval the shot is re-rendered watermark-free at 1920 × 1080 under a commercial Houdini Indie licence, with the full project file, geometry cache, PNG sequence, MP4 flipbook and Harvard-style source list delivered as one package.